

FutureXcel (Week # 3)

Data Science

Official Task Instructions:

Data Science Interns

Branding rule: Footer on every report: “Analytics by [Name]”.

WEEK 3 — Complete Exploratory Data Analysis (EDA)

What is it? Holistic investigation of dataset structure, distributions, relationships & anomalies.

What it should contain: Descriptive stats, distributions (skew, kurtosis), correlation matrix, pairwise scatter, outlier analysis, time series decomposition if applicable, 10 actionable insights.

Task description: Deliver a full EDA notebook + PDF report explaining each insight and potential next steps.

Required work:

- Python notebook with code & charts
- PDF with 10 insights + supporting visuals
- Data summary table (columns, types, missingness)

Common problems & fixes: Superficial insights → support with numbers and charts; ignoring multicollinearity → check VIF or correlation.

Recommended tools: Python (pandas, seaborn), Jupyter/Colab.

